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Conflicts of land expropriation in China during 2006–2016: An overview and its spatio-temporal characteristics

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ABSTRACT

In recent years conflicts of land expropriation in China have received a lot of concern. Recent systematic reviews highlight causes, types and resolution of land conflicts, yet very few of these studies have considered the spatial-temporal characteristics of the issue. Utilizing spatial statistical analysis and statistical software, this paper aims to build a contextual overview on Chinese land expropriation conflicts and explore spatial and temporal distribution of it during 2006–2016. Correlations of land conflict intensity with per capita GDP and urbanization rate have been studied. This paper indicates that farmers should share responsibilities for the occurrence of land conflicts as well due to their improper behavior. Offensive language and acts of defiance make the issue more complicated. Furthermore, the study reveals that the number of land expropriation conflicts in China has declined since 2013 due to government's positive actions. Overall, the conflict of land expropriation in space is categorized as "high intensity in the south and low intensity in the north". The most conflicted areas in China are south and the southwest. Among them, Guangdong and Yunnan are particular regions with a high intensity of land conflicts. Last but not least, by use of data we have shown that land conflict intensity in China does not have a statistically significant spatial pattern on the whole. However, the relationship between per capita GDP and land conflict intensity has a statistically dispersed spatial pattern.

1. Introduction

In recent years, the rapid development and urbanization has led to the constant expropriation of land in developing countries. However, this land expropriation with the public interest has frequently encountered ferocious opposition from farmers and caused sharp social contradictions (Mathur, 2013). China is one of the well-recognized countries in the world with the largest amount of land use, the most dramatic change and the most serious land conflicts (Zhong et al., 2010). Its urban construction land area increases from 6720 km^2 in 1981 to 49,982.7 km^2 in 2014, whose net growth is 43 262.7 km^2, increased by 6.44 times. The average annual growth rate is as high as 6.27% (Fang et al., 2017).

With the continuous expansion of cities in China, a wide range of serious problems such as the widening of the gap between urban and rural areas, environmental damage, sharp growth of population crises of resources and so on have been revealed. Among them, land issues have become the key problems affecting Chinese rural stability (Huang et al., 2016). In farmers' petition in China, the land issues account for more than 65%, while 73.2% of it is concerned with the expropriation of land (Liu and Sun, 2014). Land expropriation conflict becomes a threat against stability of society in China.

Meanwhile, land conflicts in China has attracted considerable empirical and theoretical analysis (e.g., Eddie et al., 2013; Hui and Bao, 2013; Tan and Zhang, 2016). Land conflict is identified by Brazilian human rights groups to include verbal confrontation, vandalism, expulsion, physical altercations and, at the extreme, assassinations (Hui and Bao, 2013). It is also taken as the peasants' struggle to change unequal agrarian social relations by Upreti (2004). Wehrmann(2008) argues that land conflict could be described as a social fact in which at least two parties are involved in order to pursue the property rights to land. Land conflict, therefore, can be understood as disputes or misuse over land property rights. Causes of conflicts in land acquisition have been studied which mainly refer to institutional flaws, illegal land acquisition by local governments, disharmony between the rapid expansion of the city and the space conversion (Zhou, 2003; Tan and Qi,

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2011; Huang and Sun, 2013). Specific analyses of land expropriation system including land property rights system, land compensation system, assessment system on the risk of land acquisition conflicts, and early warning mechanism of land acquisition conflicts are studied (Ding et al., 2013; Zhong, 2013). Different types of conflicts in land acquisition which include conflicts around the purpose of land acquisition, conflicts about the procedure of land acquisition, conflicts concerning the compensation for land acquisition and the settlement of landless peasants have been surveyed (Yu, 2005; Hou, 2015; Zhu, 2015; Liu and Peng, 2006). The research on the psychology of landless farmers and the game playing between the government and farmers are being done (Tang, 2016). However, to our knowledge there is little research on the spatio-temporal analysis of land conflicts. The characteristics of temporal and spatial distribution have not received adequate analytical discussion in the literature, despite the fact that the land acquisition conflicts frequently occur in China (Liu and Sun, 2014). In this paper, conflicts of land expropriation in China during 2006–2016 are deeply analyzed in order to build a contextual overview on Chinese land conflicts and explore the spatial and temporal characteristics of land conflicts in China. Moreover, correlation of land conflict intensity with per capital GDP and urbanization is discussed. Hopefully, it can provide guidance on land management for the decision-makers at local and regional level in China.

The structure of the paper is as follows. Methods will be introduced in next part, which is followed by the overview of land conflicts in China. Spatial and temporal distribution of land conflict is explored and the results are discussed within the national context. In the end, conclusions of the paper have been clearly stated.

2. Methods

2.1. Data collection

There are no specific governments or non-government organizations which collect the statistics concerning Chinese land conflicts until now. Land conflict cases in this article are mainly collected through the legitimate websites. Official documents, research papers and publicly available media reports are also used to verify the authenticity of the events. Taking names of county-level administrative areas, conflict of land requisition, land disputes, land acquisition violence, land conflict and so on as keywords, this paper collects and collates 106 cases of land conflicts. The time span is from June 2006 to November 2016. Obvious conflict behavior and cause of the conflict are clarified in the majority of these cases. What should be clarified is that this paper mainly discusses land conflicts in mainland China. Hong Kong, Macao and Taiwan are not included in the study areas due to the lack of relevant data. Data of GDP per capita and ratio of urbanization are from china statistical yearbook and corresponding-period databases of provincial level.

2.2. Calculation of land conflict intensity

According to the characteristics of land conflicts and survey data, the land conflict in China is divided into four types which includes tremendous land conflict, serious land conflict, considerable land conflict and ordinary land conflict.

The direct economic loss, the number of participants and casualties caused by the conflict of land expropriation are explored in each type (See Table 1). The first type is tremendous land conflict. It refers to the situation that has caused more than 10 deaths, or more than 50 serious injuries, or direct economic losses of more than 1 million yuan, or has more than 2000 people involved. The second one is called serious land conflict. Issues which result in loss of lives higher than 3 and lower than 10, or seriously injuring more than 20 and less than 50 people, or there is a direct economic loss ranging from 0.5 million yuan to 1 million yuan below in the conflict, or more than 300 people and less than 2000 people are involved in the conflict. The third one is considerable land conflict which results in loss of lives of less than 3, or seriously injuring more than 5 and less than 20 people, or direct economic loss ranging from 50,000 yuan to 0.5 million yuan, or more than 30 people and less than 300 people are involved. The fourth one is named ordinary land conflict, which results in no loss of life or less than five people being injured, or it causes a direct economic loss of up to 50,000 yuan or less, or it involves 30 people or less.

N stands for the number of persons; E stands for direct economic loss and its unit is yuan.

In this paper, land conflict intensity is used as an indicator to measure the severity of land conflict in a certain region, and it is represented by the weighted sum of the amount of land conflicts. Calculation of the formula is as following:

$$D = \Sigma\ (Kn*Cn)$$

D represents land conflict intensity in a certain region. C is the type of land conflicts. K is the value of weight of different types of land conflicts, which can be obtained by experts grading method.

3. Analyses

Statistical analyses were performed using SPSS 18.0, ArcGIS 9.3, and GeoDa 1.12. Land conflict intensity in each area was calculated first. An overview on land conflicts regarding the main subjects of land conflicts, death and loss of land conflicts, causes of land conflicts was conducted. Subsequently, time distribution was demonstrated using SPSS 18.0. A spatial analysis was done to get to find out spatial characteristics of land conflicts by the use of ArcGIS 9.3 and GeoDa software. In addition to testing for differences in absolute numbers, hierarchical cluster was used to determine the classification among block groups.

Spatial autocorrelation describes the distribution of a variable, or the distribution of the relationship between two variables, over space. The pattern of a variable can be clustered, where high (low) values occur near other high (low) values, or dispersed, or where high (low) values occur near low (high) values or the pattern could be random (Burt and Barber, 1996).

To determine if the spatial patterns of each variable are clustered, we measured the spatial autocorrelation using GeoDa software for land conflict intensity. To compare the level of a variable in one block group to that of the second variable in neighboring block groups, we conducted a bivariate analysis. This analysis determines if the spatial patterns of land conflict intensity with GDP per capital and urbanization rate are clustered, dispersed or random (Anselin et al., 2006). For example, the bivariate analysis determines if block groups with high value of land conflict intensity are next to block groups with a high or low

Table 1
Different types of land conflicts in China.

Types of land conflicts	Number of participants	Number of injuries	Number of death	Direct economic loss
Tremendous land conflict	$N \geq 2000$	$N \geq 50$	$N > 10$	$E \geq 1{,}000{,}000$
Serious land conflict	$2000 > N > 300$	$50 > N > 20$	$3 < N \leq 10$	$1{,}000{,}000 > E > 500{,}000$
Considerable land conflict	$30 < N \leq 300$	$5 < N \leq 20$	$N \leq 3$	$50{,}000 < E \leq 500{,}000$
Ordinary land conflict	$N \leq 30$	$N \leq 5$	0	$E \leq 50{,}000$

value of GDP per capital.

To determine the statistical significance of each Moran's I calculation, GeoDa calculates a pseudo p-value based on different permutations of a dataset (Anselin et al., 2006). A series of 999 different permutations with randomized versions of our dataset was run, and if the Moran's I for the observed dataset is higher than the randomized Moran's I calculations 95% of the time, the Moran's I is considered to be statistically significant ($\alpha = 0.05$) according to the calculated pseudo p-value and vice versa (Anselin et al., 2006).

4. Results and discussion

4.1. An overview of land conflicts in China

This paper collects and collates 106 cases of land conflicts with regard to subjects, time, location, reasons, and number of casualties in the land conflicts and so on. The time span of those cases ranges from 2006 to 2016. Sites of land conflicts involve 30 provinces, autonomous regions and municipalities directly under the central government. No information of Tibet autonomous region on land conflicts is reported. The main bodies of land conflicts refer to county-level governments, township level governments and land developers, which accounts for 76.9% of all subjects of land conflicts. From June 2006 to November 2016, 28 people have died and about 473 people have been injured due to land conflicts. Conflicts occur mainly in the process of land expropriation and after the land expropriation, which refers to 29 cases and 66 cases respectively, accounting for 89.6% of the total number. There are six most common reasons that can cause those land conflicts according to the statistics (See Table 2). The main reasons refer to compulsory land expropriation, farmers' discontent with land policy, improper behavior of farmers and official corruption and bribery.

Proportion of different kinds of land conflicts is known (See Fig. 1). Ordinary land conflicts accounts for 15.8%. The number of considerable land conflicts is 55.5%, accounting for more than half of the total. Serious land conflicts are responsible for 24.1%. Tremendous land expropriation conflicts take up 4.6%. Serious land conflicts and tremendous land expropriation conflicts altogether account for almost a third of the total. The proportion of land conflicts resolution in China is only 79.2%, which means that in those 106 cases of land conflicts, there are still 22 cases which are unsolved.

4.2. Time distribution of land conflicts in China

The amount of land conflicts is the most intuitive indicator of land conflicts. In this paper, the time distribution of land conflicts is analyzed based on the amount of land conflicts.

From the time distribution map of land expropriation conflicts in China (See Fig. 2), it can be concluded that the number of China's land expropriation conflicts are on the rise as a whole until the year 2013. This increase in the number of land conflicts had an upward trend since 2006. This ratio peaked in the years ranging from 2010 to 2013. After 2013, the number of land expropriation conflicts in China has declined. This could attribute to the application of reform of land requisition system in China in 2013 to some extent, when the central government transfers its focus from introduction of new land expropriation policies to the effects of land acquisition policies.

Table 2
Causes of land conflicts.

Causes of land conflicts	Cases
Compulsory land expropriation	40
Farmers' discontent with land expropriation policy	27
Improper behavior of farmers	20
Official corruption and bribery	17
unclear land ownership	1
No compensation for land expropriation	1

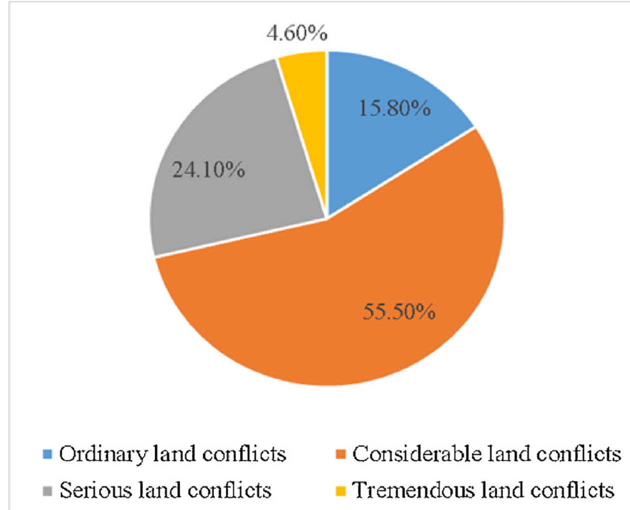

Fig. 1. Proportion of different types of land conflicts in China.

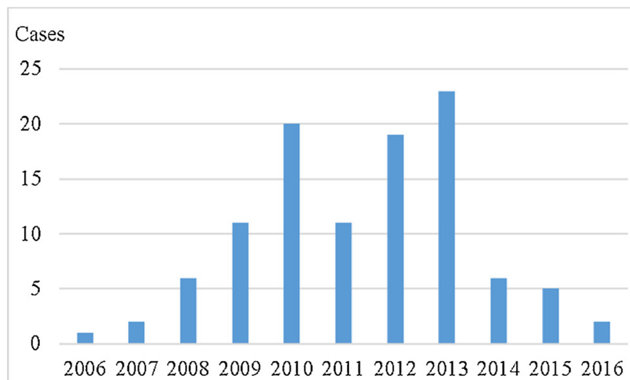

Fig. 2. Time distribution map of land expropriation conflicts in China.

4.3. Spatial distribution of land conflicts in China

A quantile map of the spatial distribution of land conflicts in 30 provinces, autonomous regions and municipalities directly under the central government is generated by ArcGIS based on land conflict intensity in each area. Hong Kong, Macao, Taiwan and Tibet autonomous region are not defined due to lack of data (See Fig. 3).

In order to understand the specific differences of land conflicts in those provinces, autonomous regions and municipalities directly under the central government, this paper adopts hierarchical clustering by using land conflict intensity of each area. The results are shown in Table 3.

The results of cluster analysis comprise 30 provinces, autonomous regions and municipalities directly under the central government which can further be divided into five categories. The first category only includes Guangdong, where value of land conflict intensity is 6.3. It is highest value of land conflict intensity, far higher than 0.81, the national average value of land conflict intensity. In the second category, Yunnan is included, which has a land conflict degree of 3.1. This land conflict intensity is relatively high. The third category contains 17 provinces, autonomous regions and municipalities directly under the central government. There are Guizhou, Hainan, Hubei, Shaanxi, Inner Mongolia, Shanxi, Shandong, Tianjin, Jilin, Jiangxi, Hunan, Qinghai, Shanghai, Chongqing, Liaoning, Xinjiang, Ningxia. The average value of land conflict intensity in those areas is 0.3 which is the lowest value of

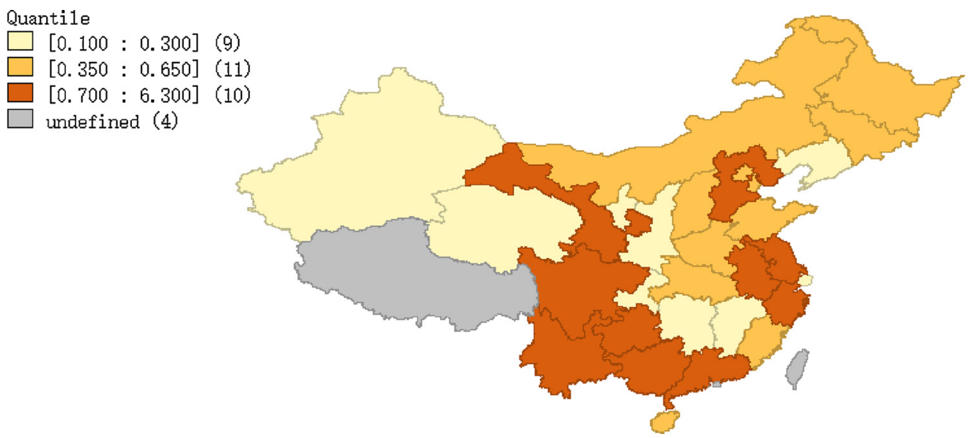

Fig. 3. A quantile map of the spatial distribution of land conflicts in China.

land conflict intensity. The fourth category includes 6 provinces and municipalities directly under the central government which refers to Gansu, Fujian, Henan, Beijing and Heilongjiang. The average value of land conflict intensity is 0.68 in those areas, which is lower than the national average value of land conflict intensity. This value of land conflict intensity is relatively low. The fifth category includes Jiangsu, Sichuan, Anhui, Zhejiang and Guangxi. The average value of land conflict intensity is 1.16 in those areas, slightly higher than the national average value of land conflict intensity. This value of land conflict intensity can be categorized as "middle".

On the whole, the conflict of land expropriation in space is characterized by "high intensity in the south and low intensity in the north". The most conflicted areas in China are in the southwest and south of China. Among them, Guangdong and Yunnan are specific regions with a high intensity of land conflicts. The northeast, northwest and north of China suffer the lowest intensity of land conflicts, where Xinjiang, Liaoning, Jilin and Inner Mongolia are the typical regions with a low intensity of land conflicts.

4.4. Spatial autocorrelation analysis

Global spatial autocorrelation analysis is carried out through Geoda software, taking land conflict intensity in 30 provinces, autonomous regions and municipalities directly under the central government as variables. The value of spatial weight is selected based on the distance. The result shows that Moran's I value is −0.009 (See Fig. 4). To assess the significance of Moran's I against a null hypothesis (no spatial autocorrelation), we use a permutation procedure, specifically the conditional permutation procedure embedded in GeoDa software. We used 9999 permutations, which in most cases are sufficient to obtain stable results (Anselin, 2003). Then, the P value of 0.39 is obtained, indicating that this spatial autocorrelation is not statistically significant (See Fig. 5). Land conflict intensity in China does not show a statistically significant spatial pattern.

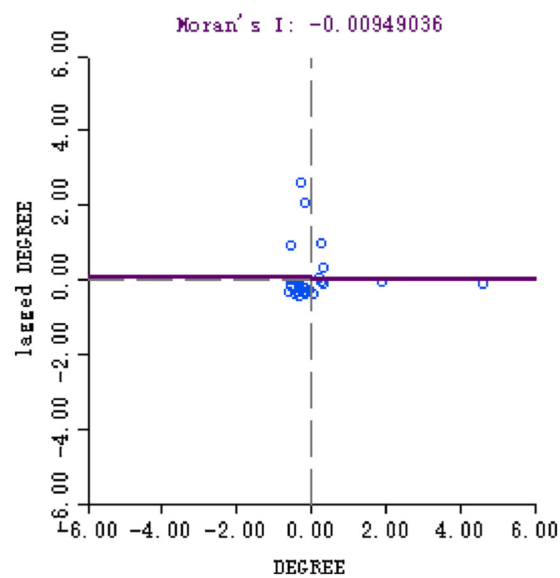

Fig. 4. Moran scatter of land conflict intensity.

4.5. Correlation with per capita GDP and urbanization rate

Per capita GDP is an important indicator used to measure regional economic development. Some studies point out that low level of economic development has led to land conflict (Li et al., 2013). In order to understand the correlation between the spatial distribution of per capita GDP and the spatial distribution of land conflict intensity, per capita GDP and land conflict intensity are selected as two variables (See Fig. 6). The spatial correlation analysis shows that Moran's I value is −0.15 and P value is 0.03. This indicates that the spatial pattern of per capita GDP and land conflict intensity are statistically significant. Since the Moran's I value is negative, a dispersed spatial pattern of each variable is suggested by the analysis, where block groups with high

Table 3
Clustering results of land conflict intensity in China.

Category	Region	Number	Mean of land conflict intensity	Order of land conflict intensity
The first category	Guangdong	1	6.3	1
The second category	Yunnan	1	3.1	2
The third category	Guizhou, Hainan, Hubei, Inner Mongolia, Shanxi, Shandong, Tianjin, Shaanxi, Hunan, Jilin, Jiangxi, Qinghai, Shanghai, Xinjiang, Chongqing, Liaoning, Ningxia	17	0.3	5
The fourth category	Hebei, Gansu, Fujian, Henan, Beijing, Heilongjiang	6	0.68	4
The fifth category	Jiangsu, Sichuan, Anhui, Zhejiang, Guangxi	5	1.16	3

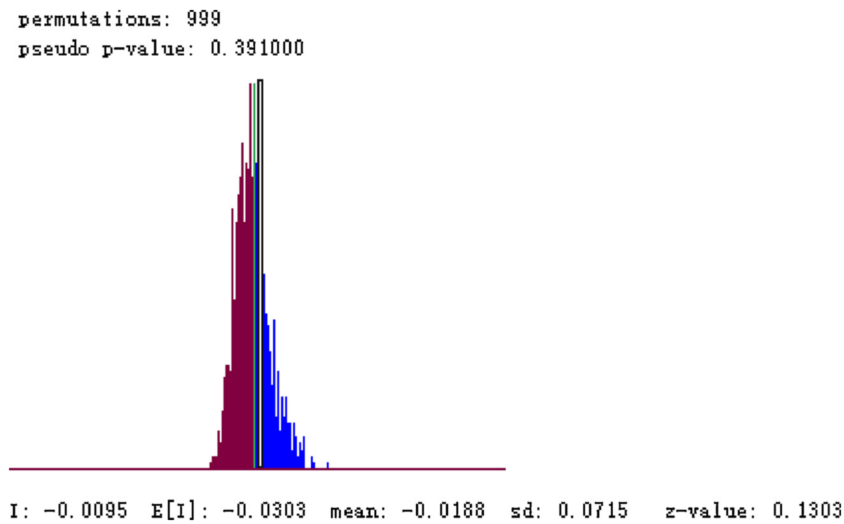

Fig. 5. Moran's I test on the spatial autocorrelation of land conflict intensity.

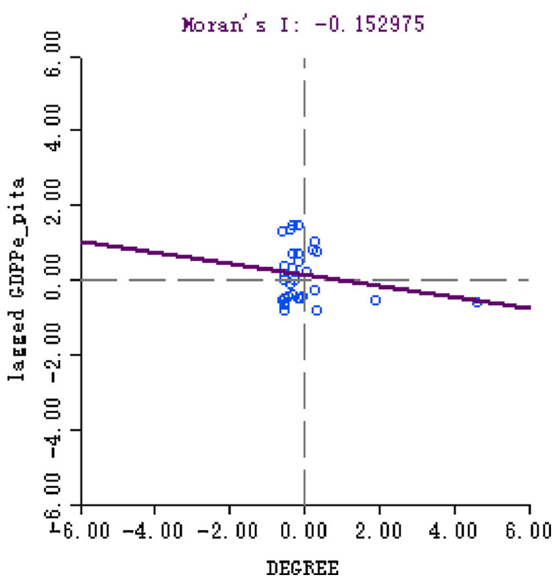

Fig. 6. Correlation distribution of land conflict intensity with per capita GDP.

value of land conflict intensity are next to other block groups with low level of economic development. Subsequently the block groups with low value of land conflict intensity are next to other block groups with high level of economic development.

Urbanization rate is the indicator used to measure urbanization. It is an important symbol of the economic development of a country or region, and a significant mark to measure the level of social organization and the level of management in a country or region (Zheng et al., 1998). Urbanization contributes to the improvement of the level of economic development (Njoh, 2003; Zhai and Jing, 2014). In this paper, the bivariate spatial autocorrelation analysis between urbanization rate and land conflict intensity is conducted. The result shows that Moran's I vale is -0.064 and p value is 0.22, which indicates that the relationship between urbanization rate and land conflict intensity does not consist of any statistically significant spatial pattern.

5. Discussion

Reasons for land conflict in China are always in hot debate. Local government' misconducts such as land expropriation with no agreement with farmers and low or even no compensation which leads to the growing discontent of farmers is always taken as the main reasons causing land conflicts (Huang and Sun, 2013). Actually, it also obtains some proofs in this study since the result shows that a majority of cases are related to it. Besides this corruption of related officials and bribery is considered to be another big reason resulting in land conflicts (Yu, 2005; Liu and Sun, 2014; Hou, 2015). Unclear land ownership also can lead to conflicts (Qian and Qu, 2004; Louis, 2001). Farmers' discontent with land expropriation policy indicates the deficiencies of land policy to some extent, which is taken as the deeper reason for land conflicts (Adam et al., 2015). Different from peer researches, our research found out that farmers also share responsibilities for the occurrence of land conflict. Improper behaviors on farmer's part such as offensive language and acts of defiance further exacerbate the conflict. Nevertheless, the root of land expropriation conflicts is the blind pursuit of personal interests (Wehrmann, 2008; Mungai et al., 2004; Hui and Bao, 2013). The game playing between local government and farmers around interests is always the critical issue.

Facing with the increases of land conflicts, Chinese local governments takes important actions to resolve the issues. Full intervention of the government plays a good role in land conflict resolution (Adisa, 2012). The coping strategies of local governments for conflict in land expropriation include legislations and regulations, policy advocacy and hearing, people's mediation and administrative mediation, administrative adjudication, reconsideration and litigation, petition and economic concessions. Moreover, a new system has been established. With the help of residents committee of the area, People Mediation Committees have been established. The committee intervenes as a third party intermediary organizations and helps solve the issue (Urresta and Nixon, 2004).

Moreover, judicial guarantee can minimize the operational risks in the process of prevention on land conflicts and uncertainty from compulsory land expropriation at utmost (McAuslan et al., 2000; Urresta and Nixon, 2004; Ding, 2007). Presently, a relatively complete legal system regarding the land acquisition procedure in China which is composed of laws, administrative regulations, departmental rules, documents by State Council and departmental normative documents have already been put in practice. These documents include Land Management Law, Regulations for the Implementation of Land Management Law, and Notice on Further Improving Land Management Work. The standardization level of land acquisition procedures is improving steadily. In addition, a series of local regulations and normative documents have been promulgated in the light of the practice of local land expropriation. The introduction of these laws provides a good system guarantee for the orderly development of land expropriation.

Together these measures facilitate the solution of land conflicts. However, it is worth noticing that the proportion of land conflicts resolution in China is only 79.2%. There are still 22 cases without any governance behavior in the data collected. This is partly because that the government is the sole buyer in rural land acquisition market and sole supplier in the urban land market in China. Lack of professional procedures to solve the land conflict in China is also an issue because many departments and organizations are involved in land expropriation. In addition, it is important to know that land conflict intensity in China does not have a statistically significant spatial pattern on the whole. It can attribute to the fast urbanization of the whole country (Tan and Qi, 2011). However, the most conflicted areas in China are in the southwest and south of China, where Guangdong and Yunnan are two particular regions with a high intensity of land conflicts. The high frequency of land conflicts in Yunnan is associated with the acceleration of the industrialization, local officials' ignorance of the legal procedure of land expropriation and the defects of land compensation systems (Li et al., 2013). Guangdong is the forerunner of reform and opening-up of China. The strong sense of rights makes it easier for citizens here to take all kinds of ways to maintain their legitimate rights and the advanced information makes the conflicts event more noticeable (Huang et al., 2006). According to the bivariate spatial autocorrelation analysis, the relationship between per capita GDP and land conflict intensity has a statistically dispersed spatial pattern, which could attribute to the industrial transfer from economically developed regions to economically backward regions. Further, we again find that the relationship between urbanization rate and land conflict intensity does not suggest a statistically spatial pattern. This provides some evidences for the fact of the universality of land conflicts in China.

It is important to note the limitations of this study. All those cases studied in this paper are mainly obtained by means of internet. We make every effort to make sure the completeness of collection of data and almost half a year is spent to collate and verify the information. Nevertheless, there is no guarantee that some of the cases on land expropriation conflicts have been covered up and are not reported which leads to the data being incomplete. However, this enumeration study represents a critical first phase in building a contextual overview on Chinese land conflicts which will support on-going efforts on the study of land issues.

6. Conclusions

Land conflict in China recently has attracted more attention of researchers. The goal of this paper is to hackle those land conflicts cases in last ten years and analyze the spatio-temporal characteristics of it. The first finding indicates that farmers should share responsibility for the occurrence of land conflicts. Use of an appropriate behavior including acts of defiance and use of offensive language only escalates the intensity of the issue. Furthermore, the study reveals that the number of land expropriation conflicts in China has declined since 2013 due to governments' positive actions. Overall, the conflict of land expropriation in space is characterized by "high intensity in the south and low intensity in the north". The most conflicted areas in China are the southwest and south of China. Among them, Guangdong and Yunnan are typical regions with a high intensity of land conflicts. Last but not least, we have explained how land conflict intensity in China on the whole does not have a statistically significant spatial pattern. However, the relationship between per capita GDP and land conflict intensity has a statistically dispersed spatial pattern.

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